Student 9 **Dara Kusumanjali** 192425398LRD 10 / 10 submitted

Q1. Sum the given number upto single digit

Score: 0.00 TC: 0/1 passed

CODE

```
n = input().strip()
while len(n)>1:
    s = 0
    for ch in n:
        s += ord(ch) - ord('0')
    n = str(s)
    print(int(n))
```

INPUT

```
89789
```

OUTPUT

```
41
5
```

Q2. Prime Number

CODE

```
def is_prime(x):
    if x < 2:
        return False
    if x == 2:
        return True
    if x % 2 == 0:
```


Q3. Swap the first and last digits

CODE

```
n = input().strip()
sign = ""
if n[0] == "-":
    sign = "-"
    n = n[1:]
if len(n) ≥ 2:
    first = n[0]
    last = n[-1]
    middle = n[1:-1]
    result = sign + last + middle + first
else:
    result = sign + n
print(result)
```

INPUT

```
23546
```

OUTPUT

```
(no output)
```

Q4. Odd Sequence first and Even Sequence next

Q4. Odd Sequence first and Even Sequence next

CODE

```
n = input().strip()
odd = []
even = []
for ch in n:
    d = int(ch)
    if d % 2 != 0:
        odd.append(ch)
    else:
        even.append(ch)
print("".join(odd) + "".join(even))
```

INPUT

```
658134
```

OUTPUT

```
(no output)
```

Q5. Step Number

CODE

```
n = input().strip()
if n.startswith('-'):
    n = n[1:]
step = True
for i in range(len(n) - 1):
    if abs(int(n[i]) - int(n[i+1])) != 1:
        step = False
        break
if step:
    print("Yes")
else:
    print("No")
```


Q6. Reverse only the even numbers

CODE

```
def reverse_even_digits(n: int) -> int:
    s = str(n)
    evens = [ch for ch in s if int(ch) % 2 == 0]
    evens.reverse()
    j = 0
    result = []
    for ch in s:
        if int(ch) % 2 == 0:
            result.append(evens[j])
            j += 1
        else:
            result.append(ch)
    return int(''.join(result))
n = 325649
print(reverse_even_digits(n))
```

INPUT

```
325649
```

OUTPUT

```
(no output)
```

Q7. Encoding

CODE

```
s = input().strip()

res = []
n = len(s)
i = 0

while i < n:
    ch = s[i]
    j = i
    while j < n and s[j] == ch:
        j += 1
    count = j - i
    res.append(ch)
    res.append(str(count))
    i = j
print(''.join(res))
```

INPUT `55553322255`

OUTPUT `(no output)`

Q8. Reverse Odd Digit

CODE

```
s = input().strip()

odd_digits = []
for ch in s:
    if int(ch) % 2 == 1:
        odd_digits.append(ch)
```


Q9. Convert every number into the next sequence number.

CODE

```
n = input().strip()

result = ""
for ch in n:
    d = int(ch)
    result += str((d + 1) % 10)

print(result)
```

INPUT `89781`

OUTPUT `(no output)`

Q10. Print the total factor count of a given number

CODE

```
n = int(input())
count = 0
for i in range(1, n + 1):
    if n % i == 0:
        count += 1
print(count)
```

INPUT `12`

OUTPUT `(no output)`